Ting Xu

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Contact Information
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Education

Washington University in St. Louis **GPA: 3.72/4.0**
Master of Science in Electrical Engineering *Aug 2021 - May 2023*
Relevant Coursework: Machine Learning, Embedded Systems, Data Science, Signal Processing

Shanghai Maritime University **GPA: 3.38/4.0**
Bachelor of Engineering in Electrical Engineering *Sept 2017 - July 2021*
Relevant Coursework: Robotics, Control Systems, Image Processing, Digital Signal Processing, Circuit Theory

Work Experience

Software Engineer, IoT and AI Systems *Mar 2025 - Present*
BHS, Inc. *Contract via Byte Carnival LLC (Mar 2025 - Sep 2026); direct employee since Oct 2026*

- Design software architecture for connected industrial equipment, turning battery room, material handling, and service requirements into device-to-cloud telemetry workflows.
- Build backend services and APIs that connect equipment data with internal enterprise systems, dashboards, and analytics services.
- Develop ingestion, validation, and transformation pipelines for sensor, equipment-status, maintenance, and operational data.
- Integrate software with embedded controllers, networked devices, and IoT gateways, accounting for latency, reliability, safety, and cybersecurity constraints.
- Develop ML workflows in Python, PyTorch, TensorFlow, and scikit-learn for predictive maintenance, anomaly detection, and operational optimization.
- Run deployment, CI/CD, containerization, logging, and monitoring for production software and analytics services.

Consultant *June 2024 - Mar 2025*
Byte Carnival LLC St. Louis, MO

- Developed innovative solutions for AI tool integration and enhanced technical workflows for global clients.
- Designed and deployed dashboards for enhanced observability and monitoring of cloud services, ensuring robust performance and reliability.
- Conducted in-depth research and analysis of client-provided evidence (STEM-related) in collaboration with a law firm, utilizing analytical frameworks to evaluate alignment with regulatory standards and strategic objectives.

Tutor *Feb 2024 - June 2024*
Varsity Tutors Inc. St. Louis, MO

- Tutored college students and professionals in STEM topics and developed tailored learning plans.
- Delivered customized lessons in Mathematics, Physics, and Computer Science.
- Designed personalized strategies to improve academic performance.

Student Worker *Aug 2021 - May 2023*
Washington University in St. Louis St. Louis, MO

- Streamlined campus-wide technology support and developed automated tools for project management.
- Streamlined routing for the tech support group by analyzing daily operations, significantly reducing travel distances and boosting campus-wide efficiency.
- Automated the pricing inquiry process for an EdTech project by developing a Python script with Selenium to systematically gather pricing data from vendor websites.

Exchange Program Coordination Intern (Volunteer) *Feb 2018 - June 2018*
International Association for the Exchange of Students for Technical Experience (IAESTE) Shanghai, China

- Facilitated cultural exchanges and internships for international students.
- Facilitated the onboarding of overseas interns and organized cultural exchange events, improving participant satisfaction.
- Matched candidates with international roles, coordinating communication across multiple stakeholders.

Awards

Scholarship for Foreign Students in Natural and Engineering Sciences *2019*
German Academic Exchange Service (DAAD)
Awarded for exceptional academic performance in engineering sciences.

Skills

Programming:	C, Python, MATLAB, C#, SQL, Shell, Git, Docker, Node.js, REST API
Machine Learning:	SciKit, PyTorch, TensorFlow, LLM, Transformers, AWS Sagemaker, YOLO
Engineering:	Embedded Systems, RTOS, MCU, DSPs, Oscilloscope, AWS IoT, I^2C, CAD, Altium
Music Technology:	MIDI, Music Theory, librosa, music21, Signal Processing, Music Information Retrieval
Research:	LaTeX, Zotero, Overleaf, Data Analysis, Data Scraping, SPSS, Manim
Languages:	English (Fluent), Chinese (Native Speaker)

Publications

1. L. Zhu, L. Zhu, Q. Wu, **T. Xu**, M. Huang, N. Gao, K. Wang, W. Wu, W. Li, C.-s. Koh, F. Blaabjerg (2025). "Design, Optimization, and Experimental Study of a Novel Direct-Driven Linear-Rotary Wave Generator." *IEEE Transactions on Energy Conversion*, vol. 40, no. 3, pp. 1712–1725.
2. L. Zhu, H. Ji, **T. Xu**, N. Gao, K. Wang, M. Huang, R. Fan, X. Jin, W. Li, C.-s. Koh, F. Blaabjerg (2025). "Three-Dimensional Analysis and Experimental Verification of a Novel Magnetic Lead Screw With Checkerboard Array Magnetic Pole." *IEEE Transactions on Transportation Electrification*, vol. 11, no. 4, pp. 8677–8688.

Research Experience

Robotic Software Research Assistant *Sept 2020 - June 2021*
ROV Research Lab, Shanghai Maritime University Shanghai, China

- Conducted research on thrust allocation and developed an optimized algorithm in MATLAB using the Singular Value Decomposition (SVD) method to improve propulsion efficiency.
- Transformed the user experience and interface for ROV control by using .NET and C# to integrate a PS4 Controller, making it easier for non-professional users to operate and improving both accessibility and innovation in how users interact.

Embedded Systems Research Assistant *July 2019 - Sept 2019*
Hochschule Emden Leer Emden, Germany

- Developed and implemented device drivers for sensors with SMBus/I^2C interfaces on the nRF52840 platform using Embedded C for Riot-OS.
- Improved the testing efficiency by automating the process through the development of a testing program.

Research Assistant *Mar 2019 - June 2023*
Department of Electrical Engineering, Shanghai Maritime University Shanghai, China

- Conducted computational optimization for a direct-driven linear-rotary wave generator (LRWG) using response surface methodology (RSM) and genetic algorithms to improve energy efficiency.
- Developed multi-objective optimization models to minimize material use and enhance system performance, emphasizing parameter sensitivity.
- Built 3-D analytical models for magnetic lead screws with Python and MATLAB, validated against finite element analysis (FEA).
- Automated data analysis workflows to enable real-time comparisons between theoretical and experimental results.

Projects

Cochlear Implant Simulator *Oct 2024 - Jan 2025*

- Implemented a Python-based tool using Librosa.
- Evaluated noise reduction algorithms for auditory enhancement.

Quiet Cool: ML for Fan Control *July 2024*

- Designed a custom ML model to balance fan noise and GPU temperature.
- Improved cooling efficiency in enterprise server environments.

Ezzy Job *Nov 2023 - Jan 2024*

- Built a backend scraper to automate the collection of job postings from Indeed by location and position.
- Developed interactive dashboards using d3.js to track job postings, providing real-time insights with enhanced UI/UX.

A Metronome Using TENS Unit *Apr 2023 - June 2023*

- Created an embedded system to stimulate muscle rhythm for better timing.
- Pioneered a novel approach in music performance technology.